

Derek Wang

+1 (416) 898-6127 | derek.wang1@uwaterloo.ca | [linkedin.com/in/4derekwang](https://www.linkedin.com/in/4derekwang) | github.com/4ppleSA0CE | derekwang.ca

EDUCATION

University of Waterloo | *Bachelor of Applied Science in Computer Engineering (Co-op)* Expected April 2030

EXPERIENCE

Software Engineer Intern May 2026 – August 2026
Prandtl Dynamics Inc. Toronto, ON

- Closed cross-tenant data exposure across 6 dashboard services with middleware that injects org-ID predicates into every query, scopes WebSocket rooms per org, and fails closed on JWT checks, enforced by CI audit gates
- Tracked 5+ DJI drones at 170 m with 20 ms latency via a real-time Python pipeline chaining SRP-PHAT localization, per-source beamforming, and Unscented Kalman Filter tracking on a single microphone array
- Raised acoustic drone detection to 98% mAP with 40% fewer false alarms at half the size via EfficientAT in PyTorch
- Automated cleaning of 50+ hours of training audio with a Python pipeline that localizes and beamforms recordings
- Slashed recording setup from 10 to 2 minutes by swapping a tethered rig for a Raspberry Pi audio service over SSH

Software Engineer Intern March 2026 – June 2026
Kenoxis Technologies Inc. Mississauga, ON

- Lowered API latency 35% by caching posting lookups and auth sessions in Redis across TypeScript/Node.js services
- Cut job-posting refresh time from daily to hourly by designing the MongoDB schemas and ingestion pipelines behind it
- Flagged suspicious applicants 60% faster by running LLM fraud screening and document review on async job queues
- Created candidate distance and map APIs with MongoDB geospatial queries to filter applicants by location

Autonomy Software Developer September 2025 – Present
WATonomous – Student Design Team Waterloo, ON

- Enabled 2 Hz obstacle-aware rover replanning via C++ ROS 2 A* on a 30x30 m LiDAR costmap and 10 Hz Pure Pursuit
- Led YOLO11s object detection retraining on 3,916 images from two merged datasets so the rover can spot a hammer, water bottle, and pickaxe at 0.96 mean average precision (mAP50)
- Accelerated YOLO11s inference to 30 FPS on the rover's NVIDIA Jetson Nano via ONNX export and onnxslim

PROJECTS

Reparo – AI Repair Assistant | Hack Canada 2026 Winner (\$5,000) | *Python, FastAPI, Swift, Gemini* [GitHub](#)

- Turned one device photo into a structured repair plan in under 60 s via component detection and Gemini reasoning
- Made every repair plan render in one format by normalizing raw Gemini output into typed FastAPI repair models
- Collapsed parts buying into one checkout with no app install via SerpAPI and Shopify APIs in a Swift App Clip

Mr. Clean – LLM-Orchestrated Mobile Manipulation | *Python, MuJoCo, PyTorch, Claude API* [GitHub](#) | [Demo](#)

- Drove a simulated two-arm robot across 3 tables from one typed prompt by building a Claude commander that hands each table to its own controller: an LLM skills agent, an imitation-learning policy, or a fly-connectome policy
- Sorted 4/4 cubes in 13 tool calls via a nested Claude agent whose 6 typed skills refuse unknown objects at the boundary
- Steered the base to 20/20 unseen goals with a policy wired from a fruit fly's brain (512-neuron FlyWire connectome)

7-DOF Robotic Arm Motion Control | *C++17, Eigen, MuJoCo, CMake, React, TypeScript* [GitHub](#)

- Wrote 7-DOF arm kinematics and dynamics from scratch in C++ using Lie groups and recursive Newton-Euler, matching MuJoCo on 1,000 test poses for forward kinematics and inverse dynamics
- Held tip error under 0.2 mm on 46/49 targets with zero actuator saturation via jerk-limited 1 kHz computed torque
- Removed arm vibration and a 2 cm tolerance by moving inverse kinematics out of the 1 kHz loop to replan on demand
- Kept the elbow clear of obstacles and joint limits while holding the tool within 0.2 mm via null-space projection

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, SQL, MATLAB

Frameworks/Libraries: React, Next.js, Node.js, Express, FastAPI, PyTorch, OpenCV, ROS 2, ONNX

Infrastructure: Linux, Docker, GitHub Actions, Git, MongoDB, Redis, SQLite, CMake